

SWR **5kWh**

High Voltage Battery



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Model	SWR5.12-H1-GS		SWR5.12-L1-GW		SWR5.12-L2-GR
MAIN INFO					
Usable Energy (Wh)	5000	Nominal Battery Energy (Wh)	5120	Usable Energy (Wh)	5000
Cell Type	LFP ($LiFePO_4$)	Usable Energy (Wh)	5000	Battery Module	SWR5.12-L2-GR: 51.2V 5.0kWh
Nominal Voltage (V)	Charge: 435; Discharge: 380	Cell Type	$LiFePO_4$	Number of Modules	1
Output Voltage (V)	320 ~ 480	Nominal Voltage (V)	51.2	Cell Type	LFP ($LiFePO_4$)
Nominal Power (W)	3000	Operating Voltage Range (V)	43.2 ~ 58.24	Nominal Voltage (V)	51.2
Peak Power	5kW, 10s	Nominal Charge Current (A)	60	Operating Voltage Range (V)	47.5 ~ 57.6
Operating Temperature Range (℃)	Charge: 0 ~ +53; Discharge: -20 ~ +53	Max. Continuous Charge Current (A)	90	Nominal Dis- / Charge Current (A)	60
Relative Humidity	0 ~ 95%	Nominal Discharge Current (A)	100	Nominal Power (W)	3000
Max. Operating Altitude (m)	4000	Max. Continuous Discharge Current (A)	100	Operating Ambient Temperature Range (℃)	Charge: 0 ~ +50; Discharge: -10 ~ +50
Communication	CAN	Pulse Discharging Current (A)	<200A (30S)	Relative Humidity	0 ~ 95%
Weight (kg)	52	Max. Continuous Charging / Discharging Power (W)	4950	Max. Operating Altitude (m)	3000
Dimensions (W × H × D mm)	700 × 380 × 170	Communication	CAN	Communication	CAN
Ingress Protection Rating	IP66	TChg (Charging Temperature Range) (℃)	0<T≤55	Weight (kg)	40
Mounting Method	Floor stacked / Wall-mounted	TDSCH (Discharging Temperature Range) (℃)	-20<T≤55	Dimensions (W × H × D mm)	442 × 133 × 420 (Excluding hanger);
Safety	IEC62619, IEC60730, VDE2510-50, CE, CEC	Ambient Temperature (℃)	40 (Recommend 10<T	Dimensions (W × H × D mm)	483 × 133 × 452 (Including hanger)
EMC	CE, RCM	Ambient Temperature (℃)	Optional heating: -20<T≤40 (Recommend 10<T≤30)	Ingress Protection Rating	IP21
Transportation	UN38.3	Relative Humidity	5 ~ 95%	Mounting Method	Cabinet / Floor stacked / Wall-Mounted
		Maximum Storage Time	12 Months (Maintenance-free)	Safety	IEC62619, IEC63056, IEC62040-1
		Max. Operating Altitude (m)	4000	EMC	EN IEC61000-6-1, EN IEC61000-6-2, EN IEC61000-6-3, EN IEC61000-6-4
		Heating	Optional	Transportation	UN38.3, ADR
		Fire Suppression	Optional, Aerosol		
		Unit Weight (kg)	50		
		Unit Dimensions (W × H × D mm)	460 × 580 × 160		
		Enclosure Protection Rating	IP65		
		Applications	On Grid / On Grid + Backup / Off Grid		
		Scalability	30P		
		Mounting Method	Wall Mounted / Grounded		
		Round-trip Efficiency*1	≥ 96%		
		Cycle Life	6000 @ 25 ± 2 ℃, 0.5 C, 70% SOH, 90% DOD		
		Safety	VDE2510-50, IEC62619, IEC62040, N140, IEC63056		
		EMC	EN IEC61000-6-1, EN IEC61000-6-2, EN IEC61000-6-3, EN IEC61000-6-4		
		Transportation	UN38.3, ADR		
		Environment	ROHS		

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SWH 3/3.6/5/6kW

Single Phase Hybrid Inverter (LV)



SWH 6/8/10/12/15kW

Three Phase Hybrid Inverter (HV)



SWH 3/3.6/5/6/8/10/12kW

Single Phase Hybrid Inverter (LV)



SWH 5/6/8/10/12/15/20kW

Three Phase Hybrid Inverter (LV)

Model	SWH3000L-S2-G2	SWH3600L-S2-G2	SWH5000L-S2-G2	SWH6000L-S2-G2
Battery Input Data				
Battery Type	Li-Ion / Lead-acid			
Nominal Battery Voltage (V)	48			
Battery Voltage Range (V)	40 ~ 60			
Start-up Voltage (V)	1			
Number of Battery Input	47			
Max. Continuous Charging Current (A)	60	75	120	120
Max. Continuous Discharging Current (A)	60	75	120	120
Max. Charge Power (W)	3000	3600	5000	6000
Max. Discharge Power (W)	3200	3600	5300	6300
PV String Input Data				
Max. Input Power (W)	4500	6300	7500	9000
Max. Input Voltage (V)	600			
MPPT Operating Voltage Range (V)	60 ~ 550			
Start-up Voltage (V)	58			
Nominal Input Voltage (V)	360			
Max. Input Current per MPPT (A)	16			
Max. Short Circuit Current per MPPT (A)	23			
Number of MPP Trackers	2			
Number of Strings per MPPT	1			
AC Output Data (On-grid)				
Nominal Output Power (W)	3000	3680	5000	6000
Nominal Apparent Power Output to Utility Grid (VA)	3000	3680	5000	6000
Max. Apparent Power Output to Utility Grid (VA)	3000	3680	5000	6000
Nominal Power at 40°C (W)**	3000	3680	5000	6000
Max. Power at 40°C (Including AC Overload) (W)**	3000	3680	5000	6000
Max. Apparent Power from Utility Grid (VA)	6000	7360	10000	10000
Nominal Output Voltage (V)	220 / 230 / 240			
Output Voltage Range (V)	170 ~ 280			
Nominal AC Grid Frequency (Hz)	50 / 60			
AC Grid Frequency Range (Hz)	45 ~ 55 / 55 ~ 65			
Max. AC Current Output to Utility Grid (A)	13.6	16.7	22.7	27.3
Max. AC Current From Utility Grid (A)	27.3	33.5	43.5	43.5
Power Factor	~1 (Adjustable from 0.8 leading to 0.8 lagging)			
Max. Total Harmonic Distortion	<3%			
AC Output Data (Back-up)				
Back-up Nominal Apparent Power (VA)	3000	3680	5000	6000
Max. Output Apparent Power without Grid (VA)	3000 (6000@10sec)	3680 (7360@10sec)	5000 (10000@10sec)	6000 (10000@10sec)
Max. Output Apparent Power with Grid (VA)	3000 (6000@10sec)	3680 (7360@10sec)	5000 (10000@10sec)	6000 (10000@10sec)
Max. Output Current (A)	13.6	16.7	22.7	27.3
Nominal Output Voltage (V)	220 / 230 / 240		220 / 230 / 240	
Nominal Output Frequency (Hz)	50 / 60		50 / 60	
Output THDv (@Linear Load)	<3%			
Efficiency				
Max. Efficiency	97.6%			
European Efficiency	96.7%			
Max. Battery to AC Efficiency	95.5%			
MPPT Efficiency	99.9%			
Protection				
DC Switch	Integrated			
DC Surge Protection	Type II			
AC Surge Protection	Type III			
AFCI	Optional			
Remote Shutdown	Integrated			
General Data				
Operating Temperature Range (°C)	-25 ~ +60			
Relative Humidity	0 ~ 95%			
Max. Operating Altitude (m)	3000 (>2000 Derating)			
Cooling Method	Natural Convection			
User Interface	LED, WLAN + APP			
Communication with BMS	CAN			
Communication with Meter	RS485			
Communication with Portal	WiFi / WiFi + LAN / 4G			
Weight (kg)	19.6	20.8	21.5	21.5
Dimension (W × H × D mm)	505.9 × 434.9 × 154.8			
Topology	Non-isolated			
Self-consumption at Night (W)	<10			
Ingress Protection Rating	IP65			
Mounting Method	Wall Mounted			

Model	SWH6000H-T1-G2	SWH8000H-T1-G2	SWH10KH-T1-G2	SWH12KH-T1-G2	SWH15KH-T1-G2
Battery Input Data					
Battery Type	Li-Ion				
Nominal Battery Voltage (V)	500				
Battery Voltage Range (V)	150 ~ 720				
Start-up Voltage (V)	150				
Number of Battery Input	1				
Max. Continuous Charging Current (A)	30	30	40	40	40
Max. Continuous Discharging Current (A)	30	30	40	40	40
Max. Charging Power (W)	9000	12000	15000	18000	24000
Max. Discharging Power (W)	6600	8800	11000	13200	16500
PV String Input Data					
Max. Input Power (W)	9600	12800	16000	19200	24000
Max. Input Voltage (V)	1000				
MPPT Operating Voltage Range (V)	120 ~ 850				
Start-up Voltage (V)	150				
Nominal Input Voltage (V)	620				
Max. Input Current per MPPT (A)	16				
Max. Short Circuit Current per MPPT (A)	24				
Number of MPP Trackers	2	2	3	3	3
Number of Strings per MPPT	1				
AC Output Data (On-grid)					
Nominal Output Power (W)	6000	8000	10000	12000	15000
Nominal Apparent Power Output to Utility Grid (VA)	6000	8000	10000	12000	15000
Max. Apparent Power Output to Utility Grid (VA)	6000	8000	10000	12000	15000
Max. Apparent Power from Utility Grid (VA)	12000	16000	20000	20000	20000
Nominal Output Voltage (V)	400 / 380, 3L / N / PE				
Output Voltage Range (V)	170 ~ 290				
Nominal AC Grid Frequency (Hz)	50 / 60				
AC Grid Frequency Range (Hz)	45 ~ 65				
Max. AC Current Output to Utility Grid (A)	8.7	11.6	14.5	17.4	21.7
Max. AC Current From Utility Grid (A)	13.7	21.0	26.1	26.1	26.1
Power Factor	0.8 leading ~ 0.8 lagging				
Max. Total Harmonic Distortion	<3%				
AC Output Data (Back-up)					
Back-up Nominal Apparent Power (VA)	6000	8000	10000	12000	15000
Max. Output Apparent Power without Grid (VA)	6000(12000@4Osec)	8000(16000@4Osec)	10000(18000@4Osec)	12000(18000@4Osec)	15000(18000@4Osec)
Max. Output Apparent Power with Grid (VA)	6000	8000	10000	12000	15000
Max. Output Current (A)	13.0 (17.4 @4Osec)	17.4 (23.3 @4Osec)	21.7 (26.1 @4Osec)	21.7 (26.1 @4Osec)	21.7 (26.1 @4Osec)
Nominal Output Voltage (V)	400 / 380				
Nominal Output Frequency (Hz)	50 / 60				
Output THDv (@Linear Load)	<3%				
Efficiency					
Max. Efficiency	98.0%	98.0%	98.2%	98.2%	98.2%
European Efficiency	97.2%	97.2%	97.5%	97.5%	97.5%
Max. Battery to AC Efficiency	97.2%	97.5%	97.5%	97.5%	97.5%
MPPT Efficiency	99.5%				
Protection					
PV AFCI3.0	Optional				
DC Switch	Integrated				
DC Surge Protection	Type II				
AC Surge Protection	Type II				
Remote Shutdown	Integrated				
General Data					
Operating Temperature Range (°C)	-35 ~ +60				
Relative Humidity	0 ~ 100%				
Max. Operating Altitude (m)	4000				
Cooling Method	Natural Convection				
User Interface	LED, WLAN + APP				
Communication with BMS	RS485, CAN				
Communication with Meter	RS485				
Communication with Portal	WiFi + LAN + Bluetooth				
Weight (kg)	23	23	25	25	25
Dimension (W × H × D mm)	496 × 460 × 221				
Topology	Non-isolated				
Ingress Protection Rating	IP66				
Mounting Method	Wall Mounted				

Model	SWH3000L-S1-G2	SWH3600L-S1-G2	SWH5000L-S1-G2	SWH6000L-S1-G2	SWH8000L-S1-G2	SWH10KL-S1-G2	SWH12KL-S1-G2
Battery Side							
Battery Type	Li-Ion / Lead-acid						
Nominal Battery Voltage (V)	48						
Battery Voltage Range (V)	40 ~ 60						
Start-up Voltage (V)	44.2						
Number of Battery Input	1			1			
Max. Continuous Charging Current (A)	70	90	120	140	160	200	240
Max. Continuous Discharging Current (A)	70	90	120	140	160	200	240
Max. Charging Power (W)	3000	3600	5000	6000	8000	10000	12000
Max. Discharging Power (W)	3300	3960	5500	6600	8800	11000	13200
PV String Input Data							
Max. Input Power (W)	6000	7200	10000	12000	16000	20000	24000
Max. Input Voltage (V)	600						
MPPT Operating Voltage Range (V)	60 ~ 550						
Start-up Voltage (V)	58						
Nominal Input Voltage (V)	360						
Max. Input Current per MPPT (A)	20	20	20	20	32 / 16	32 / 32	32 / 32
Max. Short Circuit Current per MPPT (A)	26	26	26	26	48 / 24	48 / 48	48 / 48
Number of MPPT Trackers	1	2	2	2	2	2	2
Number of Strings per MPPT	1	1	1	1	1 / 1	2/ 2	2/ 2
AC Output Data (On-grid)							
Nominal Output Power (W)	3000	3600	5000	6000	8000	10000	12000
Nominal Apparent Power Output to Utility Grid (VA)	3000	3600	5000	6000	8000	10000	12000
Max. AC Active Power (W)	3300	3960	5500	6600	8800	11000	13200
Max. Apparent Power Output to Utility Grid (VA)	3300	3960	5500	6600	8800	11000	13200
Max. Apparent Power from Utility Grid (VA)	7040	7040	8000	8800	16500	16500	16500
Nominal Output Voltage (V)	220 / 230 / 240						
Output Voltage Range (V)	170 ~ 280						
Nominal AC Grid Frequency (Hz)	50 / 60						
AC Grid Frequency Range (Hz)	45 ~ 55 / 55 ~ 65						
Max. AC Current Output to Utility Grid (A)	15	18	25	30	40	50	60
Max. AC Current From Utility Grid (A)	32	32	40	70	75	75	75
Power Factor	~1 (Adjustable from 0.8 leading to 0.8 lagging)						
Max. Total Harmonic Distortion	<3%						
AC Output Data (Back-up)							
Back-up Nominal Apparent Power (VA)	3000	3600	5000	6000	8000	10000	12000
Max. Output Apparent Power without Grid (VA)	3300 (6000, 10s)	3960 (7200, 10s)	5500 (10000, 10s)	6600 (12000, 10s)	8800 (16000, 10s)	11000 (20000, 10s)	13200 (24000, 10s)
Max. Output Apparent Power with Grid (VA)	7040	7040	8000	8800	16500	16500	16500
Max. Output Current without Grid (A)	15	18	25	30	40	50	60
Max. Output Current with Grid (A)	32	32	40	40	75	75	75
Nominal Output Voltage (V)	220 / 230 / 240						
Nominal Output Frequency (Hz)	50 / 60						
Output THDv (@Linear Load)	<3%						
Efficiency							
Max. Efficiency	97.6%						
European Efficiency	96.2%						
Max. Battery to AC Efficiency	95.5%						
MPPT Efficiency	99.9%						
Protection							
DC Switch	Integrated						
DC Surge Protection	Type II						
AC Surge Protection	Type II						
AFCI	Optional						
Rapid Shutdown	Optional						
Remote Shutdown	Integrated						
General Data							
Operating Temperature Range (°C)	-35 ~ +60						
Relative Humidity	0 ~ 95%						
Max. Operating Altitude (m)	3000						
Cooling Method	Natural Convection				Smart Fan Cooling		
User Interface	LCD, WLAN + APP						
Communication with BMS	CAN						
Communication	RS485, WiFi + LAN + Bluetooth						
Communication Protocols	Modbus-RTU, Modbus-TCP						
Weight (kg)	14.5	15.5	15.5	15.5	27.0	29.0	29.0
Dimension (W × H × D mm)	560 × 415 × 204					560 × 444.5 × 226	
Topology	Non-isolated						
Ingress Protection Rating	IP66						
Mounting Method	Wall Mounted						

Model	SWH5000L-T1-G2	SWH6000L-T1-G2	SWH8000L-T1-G2	SWH10KL-T1-G2	SWH12KL-T1-G2	SWH15KL-T1-G2	SWH20KL-T1-G2
Battery Side							
Battery Type	LFP (LiFePO4) / Lead-acid						
Nominal Battery Voltage (V)	48						
Battery voltage range (V)	40 ~ 60						
Start-up Voltage (V)	30						
Number of Battery Inputs	1	1	1	1	1	2	2
Max. Continuous Charging Current (A)	125	135	175	220	250	165	208 / 208
Max. Continuous Discharging Current (A)	125	135	175	220	250	165	208 / 208
Max. Charging Power (W)	5000	6000	8000	10000	12000	15000	20000
Max. Discharging Power (W)	5500	6600	8800	11000	13200	16500	22000
PV Side							
Max. Input Power (W)	10	12	16	20	24	30	40
Max. Input Voltage (V)	1000						
MPPT Operating Voltage Range (V)	150 ~ 850						
Start-up Voltage (V)	180						
Nominal Input Voltage (V)	620						
Max. MPPT Current (A)	20 / 20	20 / 20	20 / 20	20 / 20 / 20	20 / 20 / 20	20 / 20 / 20 / 20	20 / 20 / 20 / 20
Max. MPPT Short Circuit Current (A)	26 / 26	26 / 26	26 / 26	26 / 26 / 26	26 / 26 / 26	26 / 26 / 26 / 26	26 / 26 / 26 / 26
Number of MPPTs	2	2	2	3	3	4	4
Number of Strings per MPPT	1 / 1	1 / 1	1 / 1	1 / 1 / 1	1 / 1 / 1	1 / 1 / 1 / 1	1 / 1 / 1 / 1
AC Side (On-grid)							
Nominal Power (W)	5000	6000	8000	10000	12000	15000	20000
Max. Power (W)	5500	6600	8800	11000	13200	16500	22000
Nominal Apparent Power Output to Grid (VA)	5000	6000	8000	10000	12000	15000	20000
Max. Apparent Power to Utility Grid (VA)	5500	6600	8800	11000	13200	16500	22000
Nominal Apparent Power from Grid (VA)	5000	6000	8000	10000	12000	15000	20000
Max. Apparent Power from Grid (VA)	48300	48300	48300	48300	48300	48300	48300
Nominal Voltage (V)	400 / 380, 3L / N / PE						
Voltage Range (V)	170 ~ 290						
Nominal Frequency (Hz)	50 / 60						
Frequency Range (Hz)	45 ~ 65						
Max. Current to Grid (A)	7.9 / 8.3	9.6 / 10	12.8 / 13.4	15.9 / 16.7	19.1 / 20	23.9 / 25	31.9 / 33.3
Max. Current From Grid (A)	70						
Nominal Current From Grid (A)	7.5 / 7.2	9.1 / 8.7	12.1 / 11.6	15.2 / 14.5	18.2 / 17.4	22.7 / 21.7	30.3 / 29
Nominal Current to Grid (A)	7.5 / 7.2	9.1 / 8.7	12.1 / 11.6	15.2 / 14.5	18.2 / 17.4	22.7 / 21.7	30.3 / 29.0
Power Factor	~1 (Adjustable from 0.8 leading~0.8 lagging)						
THDi	<3%						
Back-up Side							
Nominal Output Apparent Power (VA)	5000	6000	8000	10000	12000	15000	20000
Max. Output Apparent Power (VA)	5500 (10000, 10s)	6600 (12.0, 10s)	8800(16000, 10s)	11000 (20000, 10s)	13200 (24000, 10s)	16500 (30000, 10s)	22000 (40000, 10s)
Max. Output Apparent Power with Grid (kVA)	48.3						
Max. Output Current (Bypass)	70						
Nominal Output Voltage (V)	400 / 380, 3L / N / PE						
Nominal Output Frequency (Hz)	50 / 60						
Efficiency							
Max. Efficiency	97.8%	97.8%	97.8%	97.8%	97.8%	97.9%	97.8%
European Efficiency	97.0%	97.0%	97.1%	97.3%	97.3%	97.3%	97.4%
Max. Battery to AC Efficiency	95.5%						
MPPT Efficiency	99.9%						
Protection							
Battery Reverse Polarity Protection	Optional						
DC Switch	Integrated						
DC Surge Protection	Type II						
AC Surge Protection	Type II						
AFCI	Optional						
Rapid Shutdown	Optional						
Remote Shutdown	Integrated						
General Data							
Operating Temperature Range (°C)	-35 ~ +60						
Relative Humidity	0 ~ 95%						
Max. Operating Altitude (m)	4000						
Cooling Method	Smart Fan Cooling						
User Interface	LCD						
Communication with BMS	CAN						
Communication	WiFi+LAN+Bluetooth, 4G, RS485, CAN						
Communication Protocols	Modbus RTU (RS485), Modbus TCP / IP (Ethernet), Sunspec Modbus RTU						
Weight (kg)	42.2	42.2	42.2	45.3	45.3	49.7	51.2
Dimension (W × H × D mm)	756 × 551 × 258						
Noise Emission (dB)	<65						
Power Self-consumption at Night (kW)	<15						
Ingress Protection Rating	IP66						
Anti-corrosion Class	C4						
Mounting Method	Wall Mounted						